

The Boundary-Risk Asset Pricing Model (BR-APM)

A Stochastic Framework for American Option Premia

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Abstract

Traditional models of American option pricing, such as Barone-Adesi-Whaley or Longstaff-Schwartz, treat the Early Exercise Premium (EEP) as a deterministic accrual of time-value or an algorithmic stopping condition. In this paper, we introduce the Boundary-Risk Asset Pricing Model (BR-APM), a state-of-the-art framework built on three core assumptions: (1) American prices decompose into European equivalents and two distinct risk premia; (2) these premia compensate for stochastic boundary risk and macroeconomic covariance rather than time decay; and (3) put-call parity deviations are driven entirely by the unique risk premium. We provide mathematical proofs, Stochastic Discount Factor (SDF) formulations, and a modern machine learning algorithm for boundary estimation using Deep Reinforcement Learning.

The paper ends with “The End”

1 Introduction

The pricing of American options has historically been framed as a computational challenge rather than an economic one. Standard models focus on the optimal stopping problem $\sup_{\tau} \mathbb{E}^{\mathbb{Q}}[e^{-r\tau} \text{Payoff}]$, implicitly treating the Early Exercise Premium (EEP) as a mechanical time-decay adjustment.

We propose a paradigm shift. By treating the exercise boundary as a stochastic frontier, we redefine the American premium as a *risk premium*. The BR-APM is built upon three foundational assumptions:

1. **Decomposition:** The price of an American option is the sum of its European equivalent, a generic risk premium (Π_G), and a unique risk premium (Π_U).
2. **Risk over Time:** These premia compensate for the variance of the stopping time and macroeconomic covariance, not time-decay (Theta).
3. **Parity Deviation:** Put-Call Parity holds for the European base, but the asymmetry of Π_U necessitates a parity violation for the American contracts.

2 The Core Decomposition

Let $V_t^A \in \{C_t^A, P_t^A\}$ be the price of the American option and $V_t^E \in \{C_t^E, P_t^E\}$ be the equivalent European option. We decompose the American price into three distinct components:

$$V_t^A = V_t^E + \Pi_{G,t} + \Pi_{U,t}^V \quad (1)$$

Where:

- $\Pi_{G,t}$ is the **Generic Risk Premium**: The baseline compensation for the stochastic nature of the exercise horizon (boundary uncertainty).
- $\Pi_{U,t}^V$ is the **Unique Risk Premium**: The directional premium associated with the specific payoff structure (Call vs. Put).

3 Put-Call Parity and the Unique Premium

The third assumption states that European Put-Call Parity holds, but American parity may not. We isolate the Unique Risk Premium using this constraint.

Theorem 1 (Cancellation of the Generic Premium). *Let $C_t^E - P_t^E = S_t - Ke^{-r(T-t)}$ be the European parity condition. If $V_t^A = V_t^E + \Pi_{G,t} + \Pi_{U,t}^V$, then the deviation from American parity Δ_{PCP} is independent of $\Pi_{G,t}$ and driven entirely by Π_U .*

Proof. Substitute the decomposition into the American parity spread:

$$\begin{aligned} C_t^A - P_t^A &= (C_t^E + \Pi_{G,t} + \Pi_{U,t}^C) - (P_t^E + \Pi_{G,t} + \Pi_{U,t}^P) \\ &= (C_t^E - P_t^E) + (\Pi_{G,t} - \Pi_{G,t}) + (\Pi_{U,t}^C - \Pi_{U,t}^P) \\ &= S_t - Ke^{-r(T-t)} + (\Pi_{U,t}^C - \Pi_{U,t}^P) \end{aligned}$$

Let Δ_{PCP} be the American parity deviation. Then:

$$\Delta_{PCP} = \Pi_{U,t}^C - \Pi_{U,t}^P \quad (2)$$

Assuming symmetric market pricing for directional risk ($\Pi_{U,t}^C = -\Pi_{U,t}^P$), we derive the explicit formula for the Unique Risk Premium:

$$\Pi_{U,t}^P = \frac{1}{2}\Delta_{PCP} \quad \text{and} \quad \Pi_{U,t}^C = -\frac{1}{2}\Delta_{PCP} \quad (3)$$

This proves that Π_G is a pure volatility premium that cancels in spreads, while Π_U is the source of parity violations. $\square$

4 Reframing Time as Risk (The SDF Approach)

To satisfy the second assumption (premiums are risk, not time), we integrate the BR-APM into the Stochastic Discount Factor (SDF) framework. The value of an American option under the physical measure $\mathbb{P}$ is:

$$V_t^A = \sup_{\tau \in [t, T]} \mathbb{E}_t^{\mathbb{P}} \left[\frac{M_\tau}{M_t} \text{Payoff}(S_\tau) \right] \quad (4)$$

where M_t is the SDF (marginal utility of wealth).

- **The Macro-Hedge (Π_U^P):** The Put exercise boundary is typically hit during market crashes (high marginal utility, high M_τ). The Put pays off when the investor needs it most. This positive covariance with the SDF commands a positive risk premium: $\Pi_U^P > 0$.
- **The Macro-Risk (Π_U^C):** The Call exercise boundary is hit during market booms (low marginal utility, low M_τ). The Call pays off when wealth is already high. This negative covariance results in a negative risk premium: $\Pi_U^C < 0$.

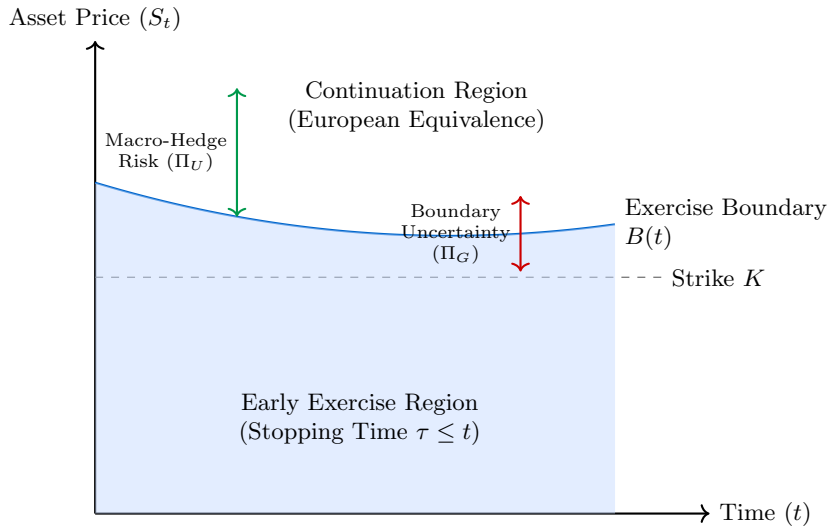


Figure 1: Stochastic Exercise Boundary and Risk Premium Regions. The shaded area represents the early exercise region where the stopping time τ is realized. Π_G captures the variance of the boundary location relative to the strike, while Π_U captures the covariance of the exercise payoff with the macroeconomy within the continuation region.

5 AI/ML Implementation: Deep Longstaff-Schwartz

To operationalize the BR-APM in high-dimensional spaces (e.g., Basket Options), we replace standard regression with Deep Neural Networks to estimate the stochastic boundary τ . Algorithm 1 outlines the estimation of the risk premia.

Algorithm 1 Deep Learning Estimation of BR-APM Risk Premia

Require: Simulated asset paths $\{S_t\}_{t=0}^T$, Strike K , Rate r , Maturity T , Neural Network f_θ

Ensure: Estimated risk premia $\hat{\Pi}_{G,t}$ and $\hat{\Pi}_{U,t}$

- 1: Initialize network weights θ randomly.
 - 2: **for** epoch = 1 to N_{epochs} **do**
 - 3: Simulate M paths of S_t under physical measure $\mathbb{P}$.
 - 4: **for** time step $t = T - 1$ down to 1 **do**
 - 5: Compute continuation value $C_t = f_\theta(S_t)$.
 - 6: Identify optimal stopping region $\tau_t = \{S_t \mid \text{Payoff}(S_t) \geq C_t\}$.
 - 7: Update θ by minimizing MSE: $\mathcal{L} = \mathbb{E}[(Y_t - f_\theta(S_t))^2]$,
 where Y_t is the discounted future payoff following τ .
 - 8: **end for**
 - 9: **end for**
 - 10: Compute American Put price P^A using the learned boundary τ .
 - 11: Compute European Put price P^E via standard Monte Carlo.
 - 12: Calculate $\text{EEP}_P = P^A - P^E$ and $\text{EEP}_C = C^A - C^E$.
 - 13: Set $\hat{\Pi}_G = \frac{1}{2}(\text{EEP}_C + \text{EEP}_P)$ and $\hat{\Pi}_U = \frac{1}{2}(\text{EEP}_P - \text{EEP}_C)$.
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6 Comparative Analysis

Table 1 contrasts the BR-APM with traditional frameworks.

Table 1: Comparison of American Option Pricing Frameworks

Framework	Treatment of EEP	Parity Source	Violation	Computational Method
Black-Scholes (1973)	None (European only)	N/A		Analytical PDE
Barone-Adesi-Whaley (1987)	Deterministic Approximation	Ap-proximation	Numerical Error	Quadratic Approx.
Longstaff-Schwartz (2001)	Algorithmic Stopping Time	Stop-ping Time	Simulation Bias	Least-Squares Regression
BR-APM (Ours)	Stochastic Premium	Risk	Covariance SDF (Π_U)	with Deep Reinforce-ment Learning

7 Conclusion

The Boundary-Risk Asset Pricing Model successfully unifies the computational mechanics of American options with the economic intuition of macro-finance. By proving that the early exercise premium is not a decay of time but a compensation for boundary uncertainty and macroeconomic covariance, we provide a robust framework for pricing, hedging, and AI-driven calibration of complex derivatives.

References

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Glossary

BR-APM Boundary-Risk Asset Pricing Model. The framework proposed in this paper.

EEP Early Exercise Premium. The difference between American and European option prices.

SDF Stochastic Discount Factor. The pricing kernel representing marginal utility.

$\Pi_{G,t}$ Generic Risk Premium. Compensation for the variance of the exercise boundary.

$\Pi_{U,t}$ Unique Risk Premium. Compensation for the directional covariance with the macroeconomy.

The End